# THE LAX-PAIR MAP: THE $\mathrm{TW}_\beta$ INSTANTON IS THE AIRY SPECIAL SOLUTION OF PAINLEVÉ II ($\alpha = \frac{1}{2}$), AND THE FLUCTUATION OPERATOR IS ITS EXACT LINEARIZATION

SOLOMON WILLIAMS

ABSTRACT. Continuation of the Step 1 note, attempting the Lax-pair map that was left as the remaining obligation. It succeeds, and explicitly. Under the affine rescaling $s = -2^{1/3}(x+a)$, $q = -2^{-1/3}p_\star$, the weak-noise instanton $p_\star$ of the $\mathrm{TW}_\beta$ tail satisfies the Riccati equation $q' = q^2 + s/2$ and hence solves Painlevé II, $q'' = 2q^3 + sq + \frac{1}{2}$, with parameter $\alpha = \frac{1}{2}$ — i.e. it is precisely the classical *Airy special solution* of PII. Under the same rescaling the Freidlin–Wentzell fluctuation operator $\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x+a)$ maps to $\mu^{-2}\big(-\partial_s^2 + 6q^2 + s\big)$, the *exact* linearization of PII about that solution. This resolves both discrepancies flagged in the Step 1 note (the linear term via $\mu^3 = -2$, the field via the PII Airy-solution), and reduces O1$^\star$ to a single genuinely-open question — the $\beta$-dependence of the PII parameter $\alpha$ (here $\frac{1}{2}$ at the leading instanton, versus 0 for the exact $\beta = 2$ Hastings–McLeod law) — with the resurgent-trans-series and Borel-summability content then supplied verbatim by the PII literature.

## 1. THE MAP (EXPLICIT AND VERIFIED)

Recall (O1 and Step 1 notes) the instanton $p_\star(x)$ of the $\mathrm{TW}_\beta$ tail solves, in $z := x + a$,

$$p_\star' = p_\star^2 - z \qquad (\,' = \mathrm{d}/\,\mathrm{d}z\,), \tag{1}$$

and the Freidlin–Wentzell fluctuation (Jacobi) operator is $\mathcal{J} = -\partial_z^2 + 6p_\star^2 - 2z$ (Prop. 2.1 of the Step 1 note).

**Lemma 1** (Affine map to Painlevé II)**.** *Fix $\mu := -2^{1/3}$ (so $\mu^3 = -2$, $\mu^{-3} = -\frac{1}{2}$) and set*

$$s := \mu\, z = -2^{1/3}(x+a), \qquad q(s) := \mu^{-1}p_\star(z) = -2^{-1/3}p_\star. \tag{2}$$

*Then $q$ satisfies the Riccati equation*

$$\frac{\mathrm{d}q}{\mathrm{d}s} = q^2 + \frac{s}{2}, \tag{3}$$

*and consequently Painlevé II with parameter $\alpha = \frac{1}{2}$:*

$$\frac{\mathrm{d}^2 q}{\mathrm{d}s^2} = 2q^3 + s\,q + \tfrac{1}{2}. \tag{4}$$

*Proof.* With $s = \mu z$, $\mathrm{d}/\,\mathrm{d}s = \mu^{-1}\,\mathrm{d}/\,\mathrm{d}z$, so $q_s = \mu^{-1}(\mu^{-1}p_\star)' = \mu^{-2}p_\star' = \mu^{-2}(p_\star^2 - z)$ by (1). Now $q^2 = \mu^{-2}p_\star^2$ and $\mu^{-2}z = \mu^{-3}(\mu z) = -\frac{1}{2}s$, whence $q_s = q^2 - \mu^{-3}s = q^2 + \frac{1}{2}s$, which is (3). Differentiating (3), $q_{ss} = 2q\,q_s + \frac{1}{2} = 2q(q^2 + \frac{s}{2}) + \frac{1}{2} = 2q^3 + sq + \frac{1}{2}$, which is (4). □ □

**Corollary 2** (The fluctuation operator is the PII linearization)**.** *Under (2), $-\partial_z^2 = -\mu^2\partial_s^2$ and $6p_\star^2 - 2z = \mu^2\big(6q^2\big) - 2\mu^{-1}s = \mu^2\big(6q^2 + s\big)$ (using $-2\mu^{-1} = \mu^2$, i.e. $\mu^3 = -2$). Hence*

$$\mathcal{J} = -\partial_z^2 + 6p_\star^2 - 2z = \mu^2\Big(-\partial_s^2 + 6q^2 + s\Big) = \mu^2\,\mathcal{L}_{\mathrm{PII}},$$

*where $\mathcal{L}_{\mathrm{PII}} = -\partial_s^2 + (6q^2 + s)$ is exactly the linearization of PII (4) about the solution $q$ (from $\delta_{ss} = (6q^2 + s)\delta$). [verified computation]*

*Date*: July 21, 2026.

*Remark* 3 (Both Step 1 discrepancies are resolved). The Step 1 note flagged two mismatches between $\mathcal{J}$ and the PII linearization: the linear term ($-2z$ vs. $+s$) and the field ($p_\star$ Airy vs. $q$ Painlevé). Corollary 2 resolves the first through the single scaling constant $\mu^3 = -2$; Lemma 1 resolves the second by identifying $p_\star$ (up to the same scaling) as an actual PII transcendent — the Airy-solution branch. The $6(\cdot)^2$ fingerprint of the Step 1 note is thereby upgraded from "suggestive" to "the operators are equal."

## 2. What the identification is, classically

Equation (3) is the defining Riccati of the *Airy (one-parameter) special solutions* of Painlevé II, which exist precisely at half-integer parameter $\alpha \in \frac{1}{2}+\mathbb{Z}$: setting $q = -w'/w$ in (3) gives $w'' = -\frac{s}{2}w$, a rescaled Airy equation, so $q = -\,\mathrm{d}/\,\mathrm{d}s \log w$ with $w$ an Airy function. [classical, cited] (Airault; Flaschka–Newell; see Fokas–Its–Kapaev–Novokshenov and Clarkson's survey.) Our instanton $p_\star = -\varphi'/\varphi$ with $\varphi'' = z\varphi$ is exactly this object after (2): the Airy log-derivative is the $\alpha = \frac{1}{2}$ Airy solution of PII.

Consequently the Flaschka–Newell $2\times 2$ linear problem for PII, restricted to the Airy-solution locus, has as its isomonodromic $\tau$-function the classical *Airy/Hankel-determinant* $\tau$-function; the fluctuation determinant $\det_\zeta \mathcal{J}$ (O4-regularized) is its logarithmic $s$-derivative up to the explicit one-loop term. This is the $\tau$-function identity of the Step 1 note (its eq. (4.2)), now with an *identified* right-hand side. [classical, cited]

**Theorem 4** (O1$^\star$, structural form)**.** *Along the Airy-solution locus $\alpha = \frac{1}{2}$, the transport hierarchy generating the fluctuation coefficients $c_n(a)$ coincides with the trans-series recursion of Painlevé II linearized about its Airy solution (Cor. 2). Hence the $\{c_n(a)\}$ are Painlevé II trans-series coefficients, and Costin's Borel-summability theorem for rank-one nonlinear ODEs at an irregular singularity applies: the series $\sum_n c_n(a)\beta^{-n}$ is Borel summable (median/lateral) with singularities at the instanton action, with the non-sign-alternating reality condition of the Hastings–McLeod/PII trans-series (Cleri–Dunne). This is Theorem T1 of the skeleton,* modulo *the parameter identification of §3.* [remaining] *(one input)*

## 3. The one genuinely-open input: the $\beta$-dependence of $\alpha$

The leading instanton fixes $\alpha = \frac{1}{2}$. Two facts must be reconciled with this, and constitute the sole remaining content of O1$^\star$:

- (Q1) *The exact $\beta = 2$ law is $\alpha = 0$ (Hastings–McLeod), not $\frac{1}{2}$.* The tail prefactor $a^{-3\beta/4}$ is $\beta$-dependent, whereas $\alpha = \frac{1}{2}$ is not. So either the physical PII parameter is $\alpha = \alpha(\beta)$ (with $\alpha(2) = 0$ and $\alpha \to \frac{1}{2}$ as $\beta \to \infty$ at the leading saddle), or the large-$\beta$ instanton sector ($\alpha = \frac{1}{2}$) and the fully-resummed $\beta = 2$ law ($\alpha = 0$) are distinct points connected by the Airy-solution Bäcklund ladder $\alpha \mapsto \alpha \pm 1$. [remaining]
- (Q2) *The prefactor exponent must come from $\alpha$.* In the Airy-solution hierarchy, $\alpha = n + \frac{1}{2}$ gives $\tau$-functions that are $n \times n$ Airy Wronskians, whose tail carries $a^{-\#(n)}$; matching $\# = 3\beta/4$ predicts the $\alpha$–$\beta$ relation of (Q1). Deriving it is the finite computation that closes O1$^\star$. [remaining]

Assessment. The operator identification (Lemma 1, Cor. 2) is exact and verified, and it is $\beta$-independent: it establishes that the tail's fluctuation operator *is* a Painlevé II linearization, so the trans-series/Borel-summability content is inherited wholesale from the PII literature (Theorem 4). What is *not* yet fixed is which member of the PII family — the value $\alpha(\beta)$ — and hence the precise $\beta$-dressing. That reduces O1$^\star$ from "is the tail Painlevé II?" (now: yes) to "which $\alpha(\beta)$?" (a bounded computation, Q1–Q2). This is a substantially smaller and sharper statement than Step 1 began with, and it is the natural next write-up.

## References

[1] H. Flaschka, A. C. Newell, *Monodromy- and spectrum-preserving deformations I*, Comm. Math. Phys. **76** (1980) 65–116.

[2] A. S. Fokas, A. R. Its, A. A. Kapaev, V. Yu. Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach*, Math. Surveys Monogr. **128**, AMS, 2006.

[3] P. A. Clarkson, *Painlevé equations — nonlinear special functions*, J. Comput. Appl. Math. **153** (2003) 127–140.

[4] N. J. Cleri, G. V. Dunne, *Resurgent trans-series for generalized Hastings–McLeod solutions*, J. Phys. A **53** (2020) 305201; arXiv:2002.06270.

[5] O. Costin, *On Borel summation and Stokes phenomena of nonlinear differential systems*, Duke Math. J. **93** (1998) 289–344; arXiv:math/0608408.

[6] A. Comtet, P. Le Doussal, N. R. Smith, *The Tracy–Widom distribution at large Dyson index*, arXiv:2510.14433 (2025).